

In OpenStack, **segregating the compute cloud** is a strategic architectural approach used to organize resources, ensure service continuity, and provide isolated environments for different users or departments 1, 2.

Purpose of Segregation

As a cloud infrastructure grows, segregation becomes necessary to:

- **Maintain Low Latency:** It helps ensure that API services remain responsive even as the number of managed resources increases 2.
- **Ensure Service Continuity:** Operators plan for segregation to handle unplanned downtime caused by natural forces or hypervisor-level failures 2, 3.
- **Redundancy:** Segregation allows for redundant service deployments to prevent a single point of failure 2.
- **Scalability:** It provides the tools necessary to manage the need for scale and high availability of compute servers 3.

Tenant Isolation Mechanisms

OpenStack employs several mechanisms to ensure that different "tenants" (users or projects) remain isolated from one another:

- **Projects (Tenants):** This is the primary method of logical segregation. Projects serve as **isolated workspaces** where resources are grouped for specific users or departments 1, 4, 5.
- **Keystone Identity Service:** Keystone acts as the central authority, managing user accounts and assigning specific **roles and permissions** within each project to ensure users only access their authorized resources 5.
- **Private Images:** In image management, images can be designated as **private**, restricting their use to specific projects or users to prevent unauthorized deployments 6.

Security Benefits

Segregation significantly strengthens the cloud's security posture:

- **Workload Isolation: Security groups** act as virtual firewalls, meticulously controlling incoming and outgoing traffic to isolate workloads from potential threats within the same cloud 1, 7, 8.
- **Architectural Shielding:** The **nova-conductor** service acts as an intermediary, ensuring that **compute nodes are denied direct access to the central database** 9, 10. This ensures that if a compute node is compromised, the attacker cannot access the core database 9.
- **Network Segmentation:** The **Neutron** service provides granular security controls and virtual network isolation, ensuring VMs cannot interfere with each other's traffic unless explicitly permitted 11, 12.

Resource Allocation Control

Segregation allows administrators to maintain granular control over how physical and logical resources are distributed:

- **Availability Zones (AZ):** Administrators group compute nodes into **fault domains** (AZs) based on physical location or power source 3, 13. This allows them to distribute VMs strategically for **fault tolerance and redundancy** 8.
- **Quota Management:** Through the **Horizon dashboard**, administrators can manage project-specific quotas, ensuring one tenant does not consume a disproportionate share of the cloud's CPU, memory, or storage 14.
- **Placement Groups:** The **nova-scheduler** respects these preferences to distribute VMs across the cloud according to specific performance or reliability needs 15.

Exam-Ready Explanation

Question: Describe the methods and benefits of segregating an OpenStack compute cloud.

- **Logical Segregation (Tenancy):** Managed via **Projects**, which provide isolated workspaces for different organizational units 1, 4.
- **Network Segregation:** Achieved using **Security Groups** (virtual firewalls) and **Subnets** to isolate traffic and protect workloads 1, 7, 12.
- **Physical Segregation (Availability Zones):** Grouping nodes into **fault domains** to ensure that a hardware failure in one zone does not impact the others 3, 8.
- **Security Pillar:** The use of **nova-conductor** as a database proxy ensures that compute nodes—the most exposed part of the cloud—never have direct access to sensitive central data 9, 10.
- **Operational Goal:** The ultimate aim is to provide **service continuity, scalability, and high availability** while maintaining **low API latency** 2, 3.